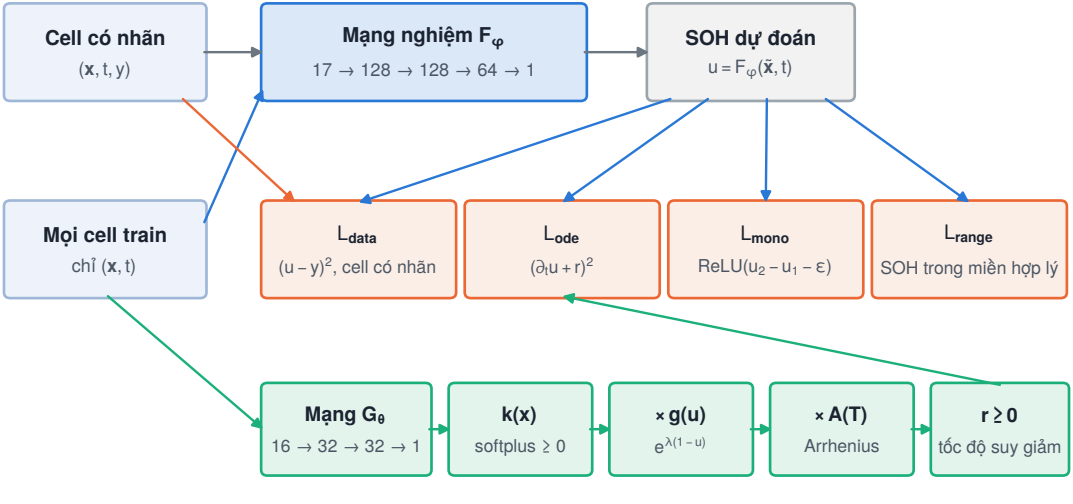

$$L = L_{data} + w(s)(\alpha L_{ode} + \beta L_{mono} + \gamma L_{range}) \quad \text{— cập nhật đồng thời } \varphi, \theta, \lambda, E_a$$